



GS FOUNDATION 2024-25

GEOGRAPHY - 8

Water in the Atmosphere - II

TABLE OF CONTENTS

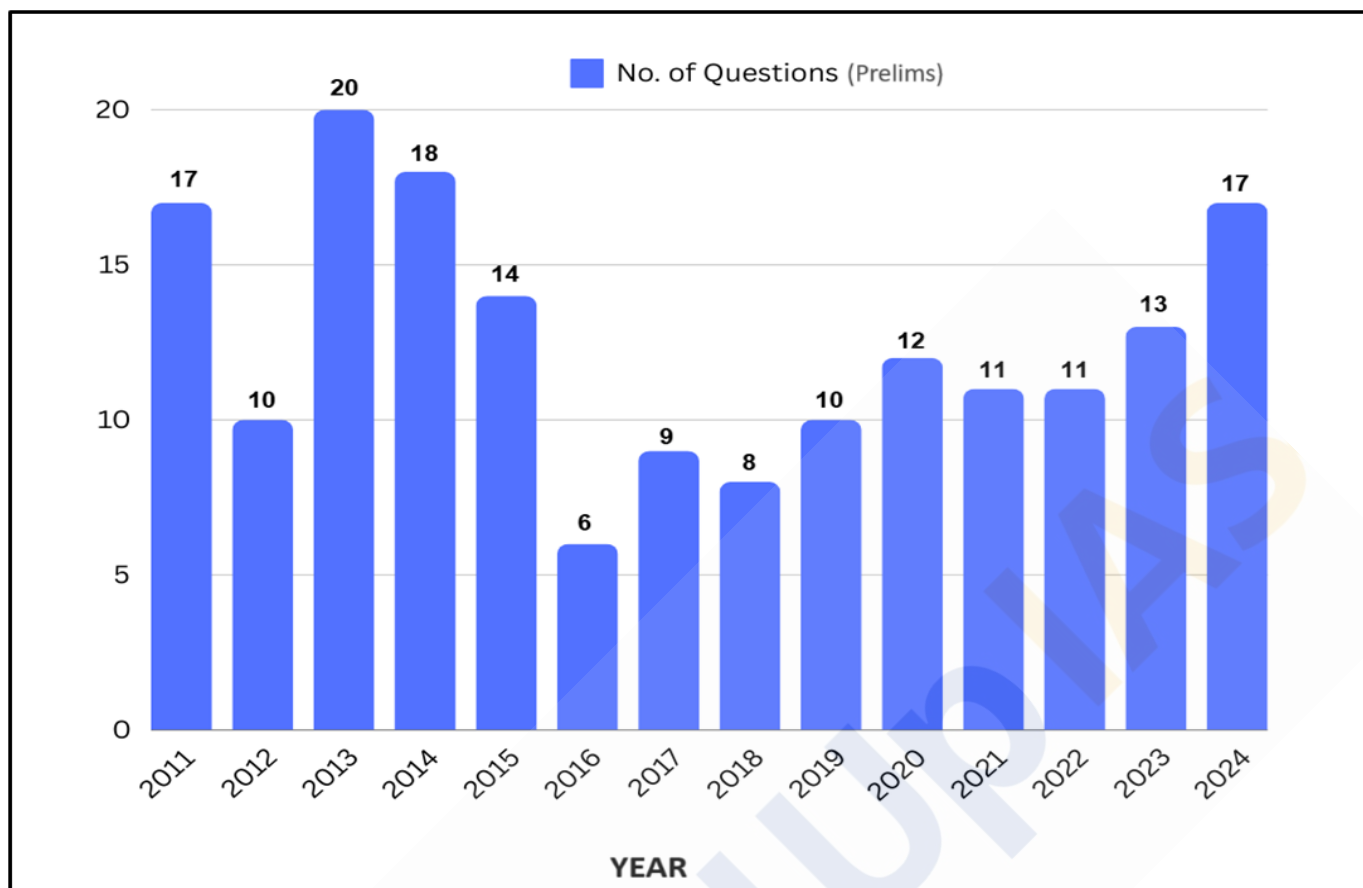
Clouds

<i>Introduction</i>	<i>3</i>
<i>Cloud Formation</i>	<i>3</i>
<i>Factors Affecting Cloud Formation</i>	<i>3</i>
<i>Formation of Clouds</i>	<i>4</i>
<i>Types of Clouds</i>	<i>4</i>
<i>Cloud Seeding</i>	<i>6</i>
<i>Methods of Cloud Seeding</i>	<i>6</i>
<i>Applications of Cloud Seeding</i>	<i>7</i>
<i>Challenges of Cloud Seeding</i>	<i>7</i>

Precipitation

<i>Introduction</i>	<i>8</i>
<i>Mechanisms of Precipitation</i>	<i>8</i>
<i>Bergeron Process</i>	<i>8</i>
<i>Collision-Coalescence Process</i>	<i>9</i>
<i>Forms of Precipitation</i>	<i>10</i>

Year-wise Trend Analysis (2011 – 2024)



Topic-wise Trend Analysis (2011 – 2024)

Physical Geography

Topic	Questions
Geomorphology	9
Oceanography	5
Climatology	22
Mapping	16
Distribution of Resources	5
Miscellaneous	15

Indian Geography

Topic	Questions
Physiography of India	24
Drainage System of India	15
Indian Climate	7
Indian Soils	5
Natural Vegetation of India	12
Indian Agriculture	31
Mineral Resources of India	10

Clouds

Introduction:

A **cloud** is a visible **mass of condensed water vapor or ice crystals suspended in the atmosphere**. These water droplets or ice crystals form when warm, moist air rises and cools. As the air cools, it reaches its dew point, the temperature at which the air can no longer hold all the moisture it contains. This excess moisture condenses around tiny particles in the air, like dust or salt, forming the droplets and crystals that make up a cloud.

Cloud Formation:

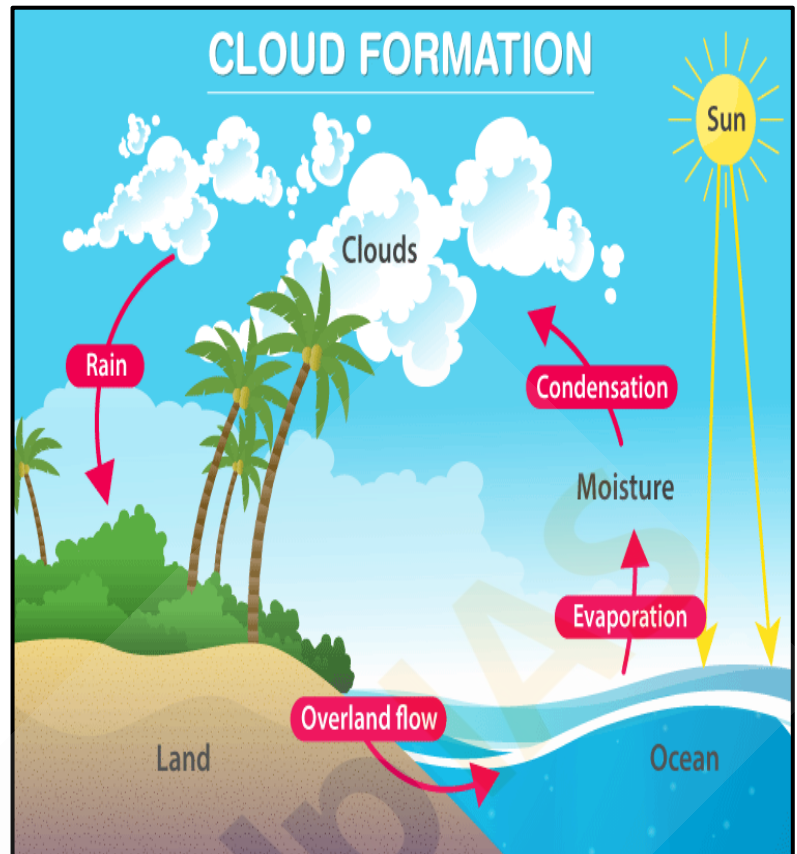
- The formation of clouds involves the **transformation of invisible water vapor into visible water droplets or ice crystals**. Water vapor and aerosol particles, such as salt and dust, are constantly present in the air, frequently colliding with each other. When the air cools, the water vapor condenses, adhering to these aerosol particles. This process continues as larger water droplets gradually form around the aerosol particles, leading to the development of clouds.
- Cloud formation typically occurs when the **air reaches a state of saturation**, unable to hold any more water vapor. This saturation, leading to **condensation**, can occur through either an **increase in the water content in the air** (such as through evaporation) **or when the air is cooled to its dew point, the temperature at which condensation initiates**. The warmer the air, the more water vapor it can hold.
- However, as the air rises and subsequently cools, its capacity to retain water vapor decreases, leading to condensation and cloud formation. The altitude at which the dew point is reached and clouds form is termed the condensation level.

Factors Affecting Cloud Formation:

- **Surface Heating** - The sun's heating of the **Earth's surface warms the air in direct contact with it**, causing it to rise. These rising air columns, often known as thermals, lead to the development of cumulus clouds.
- **Topography or Orographic Forcing** - The landscape's features, like **mountains or hills**, can influence cloud formation. When air is compelled to ascend over topographical barriers, it cools as it rises, generating layered clouds.
- **Frontal Systems** - Clouds form as a result of warm air ascending over a mass of cold, dense air along frontal boundaries. These fronts delineate the **boundary between warm, moist air and cooler, drier air**, leading to various cloud formations.
- **Convergence** - Airstreams flowing from diverse directions converge, compelling the air to ascend where they meet. This **convergence can give rise to cumulus clouds** and showery weather conditions.
- **Turbulence** - Sudden changes in **wind speed** at different altitudes can create turbulent eddies in the air, also influencing cloud development.

Formation of Clouds:

- **Rising Air** - Warm air is less dense and naturally rises. As it ascends, the pressure surrounding the air decreases, causing the air to cool down (a process called adiabatic cooling).
- **Reaching the Dew Point** - As the air cools, it reaches its dew point. This is the **critical temperature** where the air becomes saturated, and the water vapor it contains can no longer stay in its gaseous form.
- **Condensation Nuclei** - Tiny particles like **dust, pollen, or sea salt** act as nuclei for condensation, aiding in the formation of cloud droplets.
- **Cloud Formation** - As millions of these droplets or crystals accumulate, they become visible, forming a cloud.



Types of Clouds:

There is a wide range of variations in clouds in terms of height, shape, color, and the transmission or reflection of light. The main cloud genera or families are classified based on their appearance and altitude. The primary types include:

High-Level Clouds (5,000 to 13,000 meters):

- Cirrus (Ci)
- Cirrostratus (Cs)
- Cirrocumulus (Cc)

Handwritten notes:
 Cc { Cirrus
 Cs { stratus
 Ci { cumulus
 Altostratus
 Nimbostratus

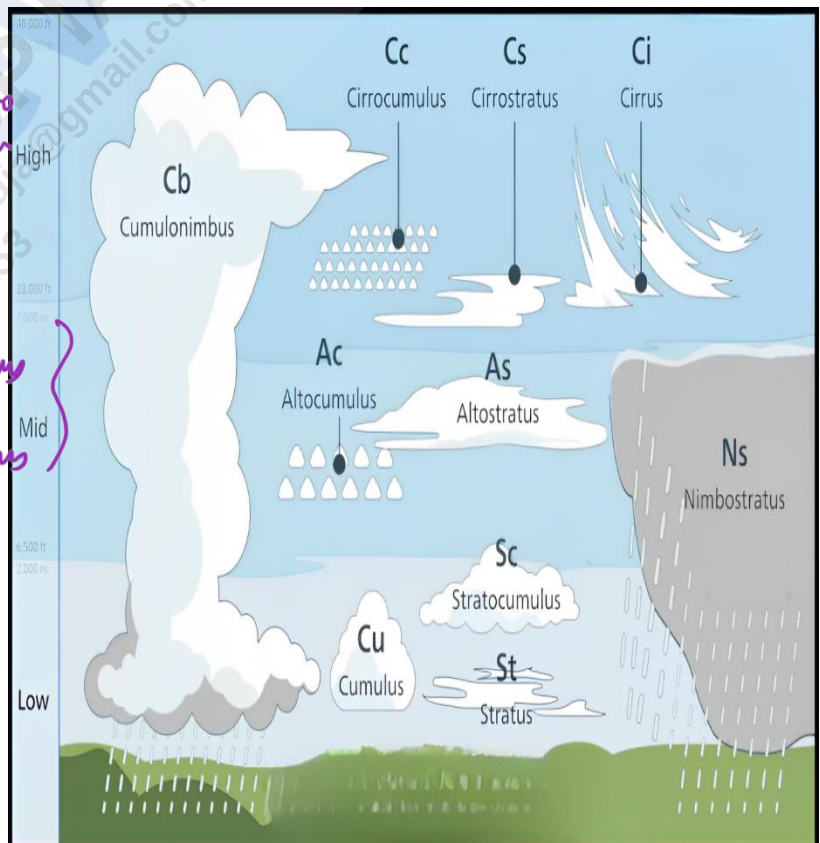
Mid-Level Clouds (2,000 to 7,000 meters):

- Altostratus (As)
- Altocumulus (Ac)
- Nimbostratus (Ns)

Handwritten notes:
 stratus
 stratocumulus
 cumulus
 cumulonimbus

Low-Level Clouds (Surface to 2,000 meters):

- Stratus (St)
- Stratocumulus (Sc)
- Cumulus (Cu)
- Cumulonimbus (Cb)



Cirrus Clouds:

- Cirrus clouds are **detached, white clouds** in the form of delicate filaments or patches. They have a **fibrous or silky sheen appearance** and are composed entirely of **ice crystals**. These high clouds are recognized for their brilliant colors at sunrise and sunset but **do not produce precipitation**. When the sun or moon shines through cirrus clouds, a **halo is often visible**.

**Cirrostratus Clouds:**

- Cirrostratus clouds are **transparent, whitish veils** that cover the sky, often creating a **halo around the sun or moon**. These high-altitude clouds are thin, giving the sky a milky appearance and are composed of **ice crystals**. **Objects on the ground do not cast shadows** when the sun is obscured by cirrostratus clouds.

**Cirrocumulus Clouds:**

- Cirrocumulus clouds appear as **thin, white patches** or layers without shading. Composed of small elements like grains or ripples, these clouds are less common and often associated with cirrus or cirrostratus clouds. They form a **'mackerel sky'** when arranged uniformly.

**Altostratus Clouds:**

- Altostratus clouds are **greyish or bluish** sheets that cover large portions of the sky. They obscure the sun or moon, which appears as a bright spot without casting shadows. These clouds consist of **supercooled water droplets** and may produce fine **drizzle or snow**.

**Alto cumulus Clouds:**

- Alto cumulus clouds are **white or grey patches, sheets, or layers** composed of rounded masses or rolls. They **do not produce halos** and often **have dark shading** on their undersides. These clouds, also known as **'sheep clouds'** or **'woolpack clouds'**, are generally found in wavy or parallel bands and can occur at various altitudes simultaneously.

**Nimbostratus Clouds:**

- Nimbostratus clouds are **dark, thick** layers that bring **continuous rain or snow**. They are low-level clouds, often thousands of feet thick, and can **blot out the sun**. Unlike other clouds, nimbostratus is not associated with **lightning, thunder, or hail**, but streaks of rain or snow, known as **'virga'**, may fall from these clouds without reaching the ground.



Stratus Clouds:

- Stratus clouds are generally **grey** with a uniform base and may produce **drizzle, ice prisms, or snow grains**. These clouds cover the sky uniformly, often without distinct form or structure. They can create a **temperature inversion** and are frequently broken, sometimes appearing similar to high fog.

**Stratocumulus Clouds:**

- Stratocumulus clouds are **low, lumpy masses** or rolls of **grey and white clouds**. They almost always have dark parts and may or may not be merged. These clouds, composed of tessellations or rounded masses, **do not produce halos**.

**Cumulus Clouds:**

- Cumulus clouds are **dense and vertically developed**, resembling **mounds, domes, or towers** with brilliant white sunlit parts and darker bases. They represent the tops of strong convective currents and produce **light precipitation**. Cumulus clouds can transition into cumulonimbus clouds, which are associated with heavier showers.

**Cumulonimbus Clouds:**

- Cumulonimbus clouds are **heavy and dense** with significant vertical extent, often forming an '**anvil head**' at the top. These clouds are associated with **severe weather, including heavy rainfall, thunder, lightning, hail, and tornadoes**. They are easily recognizable by their towering structure and sudden darkening of the sky, often leading to short-lived thunderstorms.

**Cloud Seeding:**

Cloud seeding is an artificial technique to induce rainfall by dispersing substances like dry ice or silver iodide aerosols into the upper part of clouds. The aim is to enhance the moisture content in the clouds, facilitating rain. There are three primary methods of cloud seeding: hygroscopic, static, and dynamic.

Methods of Cloud Seeding:**Hygroscopic Cloud Seeding:**

- This method disperses **salts through flares or explosives in the lower portions** of clouds. The salts attract water, growing in size and aiding in precipitation.

Static Cloud Seeding:

- Involves spreading **chemicals like silver iodide** into clouds. Silver iodide acts as a nucleus for moisture condensation.

Dynamic Cloud Seeding:

- Aims to **boost vertical air currents**, encouraging more water to pass through the clouds, resulting in increased rainfall.

Applications of Cloud Seeding:

- **Agriculture** - Cloud seeding can create rain, providing relief to drought-stricken areas. An example is 'Project Varshadhari' in Karnataka in 2017.
- **Power Generation** - Cloud seeding has been used to augment hydroelectric power production, as seen in Tasmania, Australia, over the last 40 years.
- **Water Pollution Control** - It helps maintain the minimum summer flow of rivers, diluting the impact of treated wastewater discharges from municipalities and industries.
- **Fog Dispersal, Hail Suppression, and Cyclone Modification** - The U.S.A.'s "Project Sky Water" in 1962 aimed at weather modification through cloud seeding, focusing on fog dispersal, hail suppression, and cyclone modification. During winter, cloud seeding is used to increase mountain snowpack, providing additional runoff during the spring melt season. Cumulus clouds are seeded to increase annual rainfall directly on the land.
- **Tackling Air Pollution** - Cloud seeding can potentially settle toxic air pollutants through rain. Recently, the Central Pollution Control Board and researchers considered using cloud seeding to tackle Delhi's air pollution.
- **Tourism** - Cloud seeding can transform typically dry areas into more hospitable environments, enhancing tourism.

Challenges of Cloud Seeding:

- **Potential Side-effects** - The chemicals used in cloud seeding might harm plants, animals, and people or the environment as a whole.
- **Abnormal Weather Patterns** - Artificially adding chemicals to simulate rain might change climatic patterns, causing droughts in regions that typically receive moisture.
- **Cost and Logistics** - Cloud seeding involves significant costs and logistical preparations, such as delivering and releasing chemicals into the sky via flare shots or airplanes.
- **Pollution** - Artificial rain can deposit seeding agents like silver iodide, dry ice, or salt on the ground. Residual silver from cloud-seeding projects is considered toxic. Dry ice, essentially carbon dioxide, contributes to greenhouse gas emissions and global warming.

===== XXXXX =====

Precipitation

Introduction:

Precipitation is defined as **water in liquid or solid forms falling to the earth**. It happens when continuous condensation in the body of air helps the water droplets or ice crystals to grow in size and weight that the air cannot hold them and as a result these starts falling on the ground under the force of gravity.

Mechanisms of Precipitation:

The exact process or the processes which cause the cloud droplets to join together to form large rain drops capable of falling to the earth as precipitation are not fully understood. However, two mechanisms have been proposed to explain these processes, namely, **Bergeron Process** and **Collision-Coalescence Process**. Understanding these mechanisms is essential for comprehending how precipitation forms in different atmospheric conditions.

Bergeron Process:

The **Bergeron Process**, also known as the **Ice-Crystal Theory**, was proposed by Tor Bergeron, a prominent Norwegian meteorologist, in 1933. The core of the theory is based on the concept of mechanism of the growth of raindrops.

Super-cooled Water:

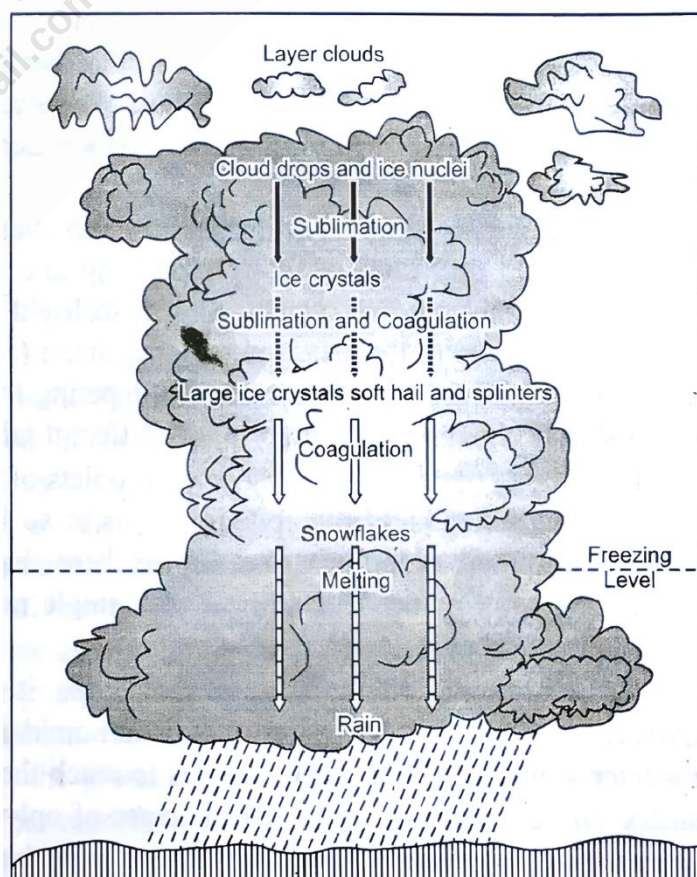
- Water droplets in clouds **do not freeze at 0°C**. In fact, super-cooled water can remain in a **liquid state even below -40°C**. This **super-cooled water** tends to **freeze if disturbed**. A typical example of this phenomenon is the icing of aircraft flying through clouds containing super-cooled droplets. Super-cooled droplets also freeze when they come into contact with tiny solid nuclei, about 1 micrometre in diameter, known as **'freezing nuclei.'** Most freezing nuclei become active at temperatures between -20°C and -25°C. However, these nuclei are sparse in the atmosphere. Consequently, as ascending air currents rise above the freezing level, some **water droplets transform into ice**, and through sublimation, water vapor converts to a solid state.

Saturation Vapor Pressure:

- Over ice**, the **saturation vapor pressure is lower than over water**. This means that when air is saturated (100 percent relative humidity) with respect to water, it is actually **supersaturated** (relative humidity greater than 100 percent) with respect to ice. For example, at -15°C, when the relative humidity is 100 percent with respect to water, it is about 115 percent with respect to ice. In this situation, vapor diffuses rapidly from the air to the ice crystals, causing them to grow at the expense of the water droplets. The **ice crystals grow quickly** enough to become large enough to **fall**.

Formation and Growth of Snowflakes:

- As ice crystals fall from the cloud, they continue to grow by intercepting cloud droplets that freeze upon them. Occasionally, these falling ice crystals break into fragments, which then act as freezing nuclei for



other water droplets, creating a chain reaction. The **ice crystals grow further through accretion**, eventually becoming **snowflakes** before leaving the cloud. Typically, snowflakes melt before reaching the ground and fall as rain. Due to the glaciation of the upper part of the cloud, its cauliflower-like top takes on an anvil shape, a distinctive characteristic of cumulonimbus clouds.

Collision-Coalescence Process:

The coalescence process of rainmaking was discovered by **E. G. Boven of Australia**. This process is applicable to clouds that do not extend beyond the freezing level. Until the 1940s, it was commonly believed that all precipitation originated through the Bergeron process, with only light drizzle coming from clouds located below the freezing point. This belief implied that substantial precipitation could only come from clouds reaching heights where temperatures were well below freezing. However, this is not accurate. In the **tropics, cumulus clouds with only 2400 m thickness can produce precipitation over oceans**, with temperatures in the upper parts of these clouds rarely falling below 7°C. Thus, ice particles contribute little to the development of rain drops in such warm clouds. Precipitation in these clouds involves the **coalescence of cloud droplets of varying sizes**. These droplets collide within the cloud, with **larger drops growing at the expense of smaller ones**. The growth rate of falling water droplets depends on variables like size, size distribution, and concentration of the drops in the cloud.

Conditions for Precipitation in Liquid Droplet Clouds:

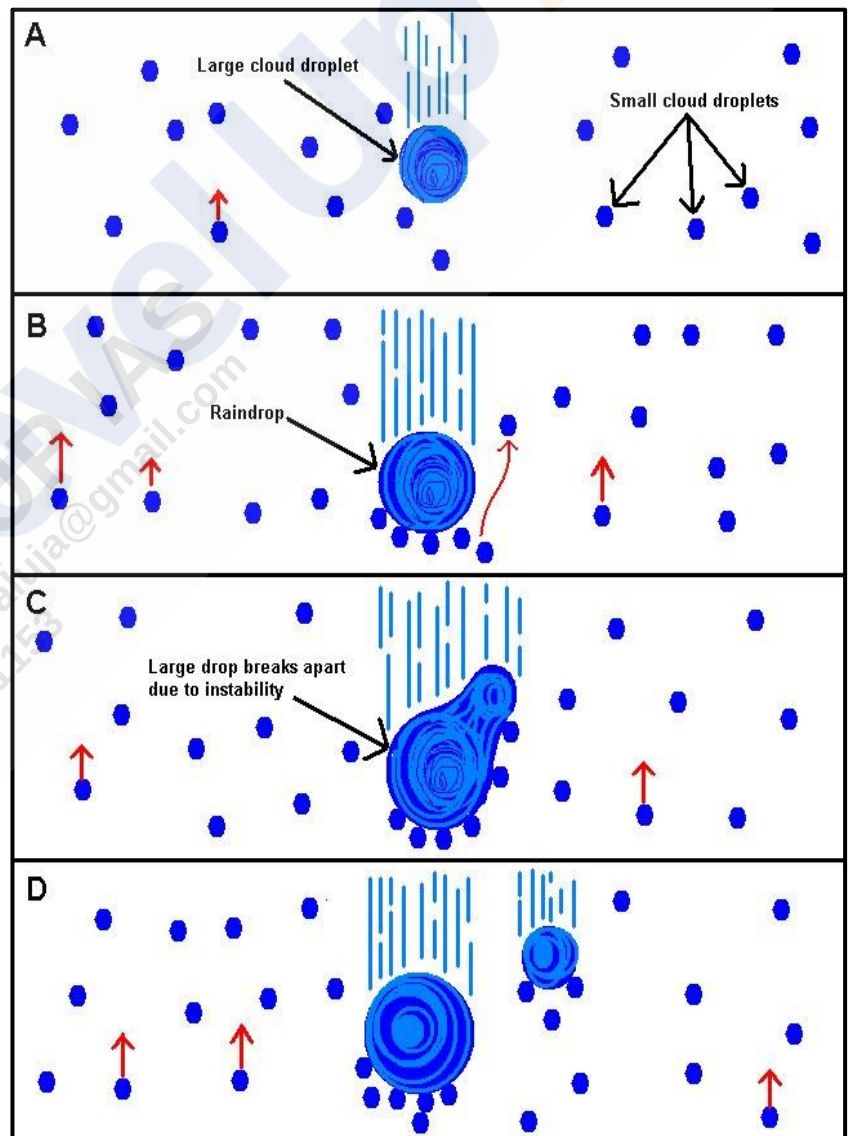
- Clouds made exclusively of **liquid droplets can generate precipitation only when they contain droplets larger than 20 micrometres**. For the formation of these large droplets, '**giant**' **condensation nuclei** are required, provided by hygroscopic materials such as sea salt. Condensation on these nuclei starts before the air reaches **saturation** (relative humidity 100%). Drops growing on such large nuclei become relatively larger, and their fall rate is size-dependent. Larger drops fall faster (or ascend more slowly in a rising cloud) than smaller ones, leading to collisions and coalescence.

Role of Vertical Air Currents and Nuclei

Size:

- In clouds with significant vertical thickness and abundant moisture, **cloud droplets are repeatedly carried upward and downward by ascending and descending air currents, quickly reaching the required size**. Two conditions aid in droplet growth:

- Initially, **few condensation nuclei** should be present to reduce competition for available water vapor.
- The **nuclei size should cover a good range to develop different fall velocities**. If a cloud contains only very small, uniform droplets, colloidal stability exists, preventing precipitation, as all droplets descend slowly at



a uniform rate with little chance of collision. Developing a single raindrop requires about one million cloud droplets to coalesce.

Required Size of Raindrops:

- **Raindrops must be over 100 microns in diameter** in humid air. A 500-micron drop would take about 10 minutes to fall from a cloud base 100 meters above ground, while a 10-micron drop would take at least 50 hours. **Drops below 500 microns are considered 'drizzle.'** Most cloud droplets are too small to fall and would evaporate before reaching the surface.

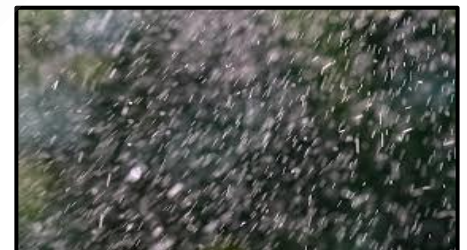
Role of Electrical Attraction in Coalescence:

- High-speed motion pictures have demonstrated that mere collisions do not always lead to coalescence. **Electrical attraction between droplets plays a crucial role in coalescence.** If colliding droplets have opposite electrical charges, coalescence is more easily achieved.

Forms of Precipitation:

The precipitation falls on the earth in various forms of droplets of water, ice flakes and solid ice balls or hail and at times droplets of water and hail together. The form that precipitation takes is largely dependent upon the method of formation and temperature during the formation. The forms of precipitation are as follows:

- **Rain** - It is the most widespread and important form of precipitation, occurring when there is **sufficient moisture and condensation above 0°C**. Rain consists of liquid water droplets larger than 5 mm in diameter, with each raindrop containing about 8 million cloud droplets. Raindrops fall at a velocity 200 times greater than cloud droplets. When they become very large, they fall faster (over 30 km/h) and may split, causing heavy downpours. Sometimes, **rain falls from cumulus clouds at higher altitudes**, but the streaks evaporate before reaching the ground, known as virga.
- **Drizzle** - Drizzle is a **fairly uniform precipitation composed exclusively of fine drops of water** with diameter less than 0.5 mm. Only when droplets of this size are widely spaced are called rain.
- **Snowfall** - When condensation takes place below freezing point (-0°C), the water vapour changes into **tiny ice crystals**. These tiny ice crystals grow in size and form ice flakes which become big and heavy and start falling on the ground. This form of precipitation is called **snowfall**. Snowfall is very common in Western Himalaya and mid and high latitude regions in winter.
- **Sleet** - Sleet is **frozen rain**, formed when rain before falling on the earth, passes through a cold layer of air and freezes. The result is the creation of solid particles of clear ice. It's usually a combination of small ice balls and rime.



- **Hail** - Hail is precipitation of **small balls or pieces of ice** (hail stones) with diameters ranging from 5 to 50mm, falling either separately or agglomerated into irregular lumps. Hailstones are comprised of a series of alternating layers of transparent and translucent ice.



===== XXXXX =====